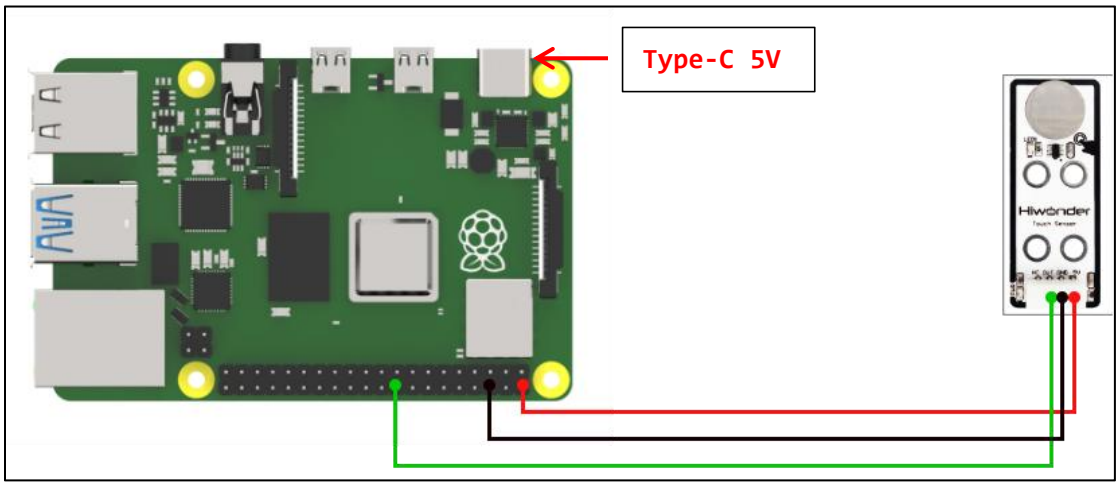
Shenzhen Hiwonder Technology Co,Ltd

Hiwonder Raspberry Pi Development Tutorial V1.0

1. Getting Ready

1.1 Wiring Instruction

This section uses DuPont wire to connect the touch sensor as an example. The wiring method is shown below:



Note:

① If you are using the lithium battery provided by our company, please connect its red wire to the positive terminal (+) of the DC port, and its black wire to the negative terminal (-).

② If the lithium battery wires are not connected, do not directly connect them to other wires to avoid a short circuit between the positive and negative poles, which may cause damage to the device.

1.2 Environment Configuration

To install NoMachine on your PC, you can access the software package in “2. Software Tools&Program Collections->01 Software Installation Packages-> Remote Desktop Connection Tool”. For detailed instructions on how to use NoMachine, please refer to the corresponding directory.

Drag the program and the library file “SDK” into the Raspberry Pi system mirror image. Take placing them on the desktop as an example. (Note: Place

the library file and the program in the same directory.)

Open the terminal, and input the command “sudo chmod a+x Sensor_Demo/” to switch to the directory where the program is.

```
ubuntu@ubuntu:~/Desktop$ sudo chmod a+x Sensor_Demo/
ubuntu@ubuntu:~/Desktop$
```

2. Case Test

This case uses the Raspberry Pi terminal window to display the detecting status of the touch sensor.

2.1 Program Download

1) Open the terminal, and input the command “cd Desktop/Sensor_Demo/” to switch to the directory where the program is. Press “Enter”.

```
ubuntu@ubuntu:~$ cd Desktop/Sensor_Demo/
ubuntu@ubuntu:~/Desktop/Sensor_Demo$
```

2) Enter the command “python3 TouchSensorDemo.py” to execute this case program.

```
ubuntu@ubuntu:~/Desktop/Sensor_Demo$ python3 TouchSensorDemo.py
```

2.2 Program Outcome

Use the touch sensor to test the touch state. When you do not touch the metal surface and the sensor does not detect any touch, the terminal window displays 1. When you touch the metal surface with your finger, the sensor detects the touch and its LED1 lights up, and the terminal window displays 0.

```
OUT: 1
OUT: 1
OUT: 1
OUT: 1
OUT: 0
OUT: 0
OUT: 0
OUT: 0
OUT: 0
```

2.3 Brief Program Analysis

- **Import the Library File**

```
import os
import sys
import time
import RPi.GPIO as GPIO
```

Import the necessary library files for the program, including system libraries, libraries required for the delay, and GPIO libraries.

- **Initialization Operation**

```
GPIO.setwarnings(False)
GPIO.setmode(GPIO.BCM)
```

Initialize GPIO port to set GPIO mode to BCM mode.

- **Main Function**

```
while True:
    GPIO.setup(24, GPIO.IN)      #Set pins to input mode
    state = GPIO.input(24)       #Read pin values
    print('OUT:',state)

    time.sleep(0.2)
```

In the “while” loop, set the GPIO pin24 to input mode. Then, store the signal value input by the pin24 in the “state” variable.

Next, print the value of the “state” variable on the terminal window.